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Research paper

Ain't blaming you: Delegation of financial decisions to humans and algorithms

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ABSTRACT

This article investigates the tendency to prioritize outcomes when evaluating decision-making processes, particularly in situations where choices are assigned to either a human or an algorithm. In our experiment, a Principal delegates a risky financial decision to an Agent, who can choose to act independently or to use an algorithm. The Principal then rewards or penalizes the Agent based on investment performance, while we manipulate the Principal's knowledge of the outcome during the evaluation. Our results confirm a significant outcome bias, indicating that the assessment of decision effectiveness remains heavily influenced by results, whether the decision is made by the Agent or delegated to an algorithm. Furthermore, the Agent's reliance on the algorithm and the level of investment risk do not change depending on whether rewards or penalties are decided before or after the outcome is known.

1. Introduction

In real-world scenarios where uncertainty plays a prominent role, it is intricate to establish straightforward cause-and-effect relationships between actions and results. In such cases, individuals often assess decisions retrospectively, considering the decision-making process and the eventual outcome. For example, witnessing Steven Bradbury on the winner's platform in short-track speed skating at the Salt Lake City Olympics could mislead someone into assuming he was the swiftest skater worldwide. He happened to be trailing behind throughout the race, and the preceding four competitors stumbled in the last turn while vying for the lead. Judging the performance from the outcome would have been misleading for a casual observer.

Under uncertainty, the quality of the decision-making process should depend only upon information available to the decision-maker when deciding (ex-ante). Thus, the outcome of the decision should not matter when assessing the quality of a decision. However, the tendency to evaluate decisions mainly on their outcome (ex-post) is a pervasive phenomenon affecting the evaluation of decisions in a wide range of contexts (Baron & Hershey, 1988). This biased assessment of the quality of the decision process is known in the literature as *outcome bias* and might lead to confirming the chosen strategy when the outcome, defined by chance, is favorable, even if the decision is wrong (Lipshitz, 1989). Agrawal and Maheswaran (2005) study the psychological mechanisms underlying the outcome bias and show that the outcome bias is largely driven by the desire to preserve one's self-image. The key role of emotional factors in the outcome bias is also confirmed by Ratner and Herbst (2005), with people being less likely to choose an option that is otherwise perceived as favorable after a bad outcome.

The outcome bias is widespread and manifests itself in various domains like human resource management (Bankins et al., 2022), sport (Lefgren et al., 2015), and medical decision-making (Sacchi & Cherubini, 2004). Here, our attention goes to choices having financial content, a domain in which the outcome bias is a well-documented phenomenon. For example, Pelster and Breitmayer (2019) show that on a social trading platform, the likelihood of obtaining likes or copiers increases in the other trader's past performance and is unaffected by risk-taking. Germann and Weber (2018) provide experimental support to the robustness of the outcome bias in financial decisions, even in situations in which the contribution of sheer luck is evident. In the experiment by König-Kersting et al. (2021), a Principal could reward an Agent who invested resources on their behalf, either *before* knowing the outcome or *after* knowing it. The authors find that rewards given after knowing the outcome are generally higher for successful investments than for unsuccessful investments, while rewards given before knowing the outcome are in between. Casal et al. (2019) allow a Principal to communicate the desired investment level to an Agent, who, however, gets the maximum reward when investing all the Principal's resources. The experiment shows that deviations from the desired investment level get punished only when the investment is unsuccessful. Gillenkirch

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and Velthuis (2023) highlight the potentially harmful consequences of outcome bias in decision-making for others: as clients' assessments become increasingly outcome-driven, decision-makers are more likely to make choices that do not serve their clients' best interests.

In line with this literature, we adopt a normative stance when assessing the outcome bias, assuming that judging actions based on their actual outcomes deviates from the moral standard of evaluating them based on their potential outcomes. This perspective is grounded in a consequentialist approach, which characterizes expected utility theory—the dominant framework in economics. However, it is important to recognize that alternative normative frameworks may also coexist. For instance, the evaluation of risky choices may arise from a normative judgment in which people apply an *actualist* utilitarian approach, assessing the utility of the outcome that ultimately materializes (for an illustration of different utilitarian approaches to risk, see Hansson, 2023). Empirical support for this perspective comes, for example, from Max and Uhl (2023) who investigate moral luck in investment contexts and found that people consciously perceive unprofitable investments as less moral. Kataria et al. (2014) experimentally study the impact of restricting choice freedom in risky decision-making. They find that such restrictions elicit a negative reaction from those affected when the limitation results in a worse outcome compared to a condition without restrictions. Conversely, when the restriction leads to a better outcome, the reaction is positive.

In the current paper, we present novel evidence about the outcome bias in the context of experimental financial decisions that can be delegated to a human or an algorithm. Specifically, we investigate whether the outcome bias is more pronounced when investment choices are made by humans or by an algorithm. The difference may occur due to the potential for algorithms to obscure the direct link between Agent intentions and investment outcomes, which could influence how Principals attribute responsibility. Given the increasing use of algorithms in supporting human decision-making, this seems to be a relevant question. Furthermore, we investigate whether algorithms are adopted strategically to shift responsibility for the investment outcome.

An important aspect of our setting is choice delegation, i.e., giving up the right to choose and letting another decision-maker make a choice that concerns us (e.g., Füllbrunn et al., 2020). Several studies highlighted how delegation to other humans is used strategically to shift responsibility for the outcome of a decision (Chang et al., 2016). The literature has mostly focused on delegation to other humans, but recently, attention has been paid to delegation to algorithms. For example, Feier et al. (2022) show that delegation to an algorithm is more effective in deflecting potential retaliation by unhappy counterparts than delegation to a human. Hohenstein and Jung (2020) confirm that AI can serve as a "moral crumple zone" and "absorb" some of the responsibility of the human decision-maker. Dana et al. (2007) explore the motivations to delegate to an algorithm and shows that delegation to an algorithm is deliberately adopted to preserve one's self-image in case of moral dilemmas.

Delegating decisions to algorithms is a widespread phenomenon in many domains of human activity involving computational tasks (Walsh & Anderson, 2009). For example, algorithms are largely employed to support decision-making in healthcare (Grote & Berens, 2020). In the financial sector, reliance on robo-advisors is growing, even though actual undertaking by investors is still scattered (D'Acunto et al., 2019). Holzmeister et al. (2022) in their experiment demonstrate that humans tend to choose investment algorithms over finance professionals. At the same time, Germann and Merkle (2022) show that humans exhibit algorithm aversion when they perceive the algorithms to be opaque, complex, or have low accuracy. Furthermore, mistakes made by the algorithm are more severely judged than mistakes made by humans, representing an impediment to the diffusion of the technology (Dietvorst et al., 2015). As of today, evidence about the consequences of algorithms in the financial sector is mixed. For example, D'Acunto et al. (2019) show that adopting robo-advisors promotes better portfolio diversification. In contrast, Asparouhova et al. (2020) highlight how humans tend to over-rely on autonomous robots, leading to poor trading outcomes.

In our experiment, an Agent has to make a risky financial decision for a Principal. The former can choose how to invest firsthand or delegate to an algorithm. The Principal, in turn, rewards or punishes the Agent. The main controlled manipulation in the experiment is the information available to the Principal when rewarding the Agent: the Principal chooses the reward either before knowing the outcome of the investment (*Reward Before*) or after (*Reward After*). Results confirm the presence of outcome bias, with the highest rewards in *Reward After* for successful outcomes, followed by *Reward Before*, and *Reward After* for unsuccessful outcomes. Furthermore, we find that the outcome bias does not differ when a choice is made personally or delegated to the algorithm. Finally, delegation to an algorithm and the level of risky investments do not differ across conditions (*Reward Before*) and (*Reward After*).

The evidence collected here contributes to the growing literature on outsourcing investment decisions to automated algorithms and robo-advisors. We demonstrate that the outcome bias is robust and does not depend on the source of the investment choice. Thus, outcome bias is not mediated by delegation to an algorithm. Previous research found that principals judge the same decision differently despite their exact knowledge of the investment strategy (König-Kersting et al., 2021). In our study, we go further, showing that artificial agents generate the same pattern of outcome-based evaluations as humans. We also contribute to the literature on shifting blame to others (e.g., Bazerman & Gino, 2012). In our experiment, the Agent does not delegate more to the *Reward after* algorithm than the *Reward before* algorithm, nor do they change their risk exposure. Thus, differently than what was previously found (e.g., Leib et al., 2021), our agents seem to not "hide behind the machine".

2. Methods and materials

We present the main components of our pre-registered experimental design and the hypotheses we are testing, with details on the experimental design, hypotheses, and analysis plan.[1]

2.1. Decision setting

We adopt a modified version of the investment game as our main decision task (Gneezy & Potters, 1997). In our investment task L, an Agent must decide the share s of their endowment E allocated to a risky asset (*Asset A*) whose return is +250% with probability 1/3 (Success) and −100% with probability 2/3 (Fail). The endowment not allocated to Asset A goes automatically into a risk-free *Asset B*, which delivers no return. Given this, the expected return of the investment task L is $E[L] = \frac{3.5}{3} sE + (1 - s)E$. The expected return of investing in *Asset A* is thus 16.7%. For example, if s=.5, the expected outcome of the investment is 108.3.

The share invested s can be taken as a direct measure of the risk propensity of the investor, with extremely risk-averse subjects investing none of their endowment into the risky asset (s=0) and risk-neutral/seeker investing all their endowment (s=1). All s in between capture different degrees of risk-aversion, with larger s associated with lower risk-aversion. To improve comparison across experimental conditions, we limited the options available and imposed $s \in \{0, .25, .50, .75, 1\}$.

In our experimental setting (see Fig. 1), two individuals interact in the task L: one is randomly assigned to the role of Agent (A) and the other to the role of Principal (P). A faces a choice about the investment of E on behalf of P in task L. Specifically, A can choose between two alternatives:

[1] The preregistration is available at https://osf.io/3uf7s.

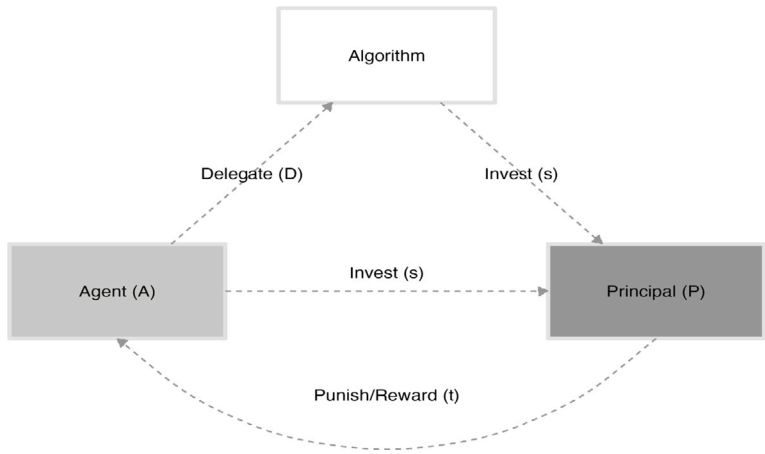

Fig. 1. Interaction between the Principal and the Agent.

- *No delegation*: *A* decides how to invest *E* firsthand;
- *Delegation*: *A* delegates the investment decision to a pre-programmed algorithm.

If *A* chooses *No delegation* ($D = 0$), the share *s* to be invested in the risky asset is freely chosen by *A*, and the payoff of the investment, which goes to *P*, is given by the return function above. If *A* chooses *Delegation* ($D = 1$), the share *s* is randomly drawn from the distribution of choices made by humans in a previous independent study, that we label *Preliminary* (see Section 2.4 for a description). The agents are told that "*the algorithm will select an investment level in Asset A according to a programmed method based on Agent choices in a previous independent session.*" The working of the algorithm is not further specified to mimic the opaqueness of real-world algorithms assisting decision-making. Thus, although our algorithm is de facto a simple random draw from a distribution of human-generated choices, participants are unaware of its working.

The payoff of the P (π_P) is given by the return of the investment task (L), both when A delegates to the algorithm and when they do not. A's payoff (π_A) is fixed. It does not depend on choices made about delegation and task *L*.[2] Furthermore, P can reward or punish A for their choice of s at no cost to themselves. Specifically, P can attribute points (t) to A. The points span over a range $t \in \{-50, -49, \ldots, 49, 50\}$, with negative points subtracted from A's payoff (punishment) and positive points added to A's payoff (reward).

2.2. Treatments

Our main experimental manipulation concerns the knowledge of the investment outcome when the Principal chooses to reward/punish the Agent. Specifically, when making this choice, the Principal always knows the investment level in the risky asset (s) and whether the Agent or the algorithm (D) made the investment choice. The Principal may not know the investment outcome π_P. Accordingly, we implement two treatments that refer to the Principal's knowledge of the investment outcome when choosing to reward/punish the Agent:

- *Reward After*: When choosing the punishment/reward (t), the Principal is aware of the investment share (s) and the delegation choice (D), and *knows* the investment outcome π_P;
- *Reward Before*: When choosing the punishment/reward (t), the Principal is aware of the investment share (s) and the delegation choice (D), but *does not know* the investment outcome π_P.

The controlled manipulation aims at testing whether the Principal's knowledge of the investment outcome affects the choice of t. In the *Reward Before* treatment, the Principal may choose t to reward or punish the Agent conditional upon the investment level s and the delegation choice D. Differently, in the *Reward After* treatment, the Principal may choose t to reward or punish the Agent based on the investment level s, the delegation choice D, *and* on the investment outcome π_P.

2.3. Hypotheses

Section 1 highlighted the relevance of the outcome bias for several domains of economic decision-making (Agrawal & Maheswaran, 2005; Ratner & Herbst, 2005), including financial choices (Germann & Weber, 2018; Pelster & Breitmayer, 2019) and decision making for others (König-Kersting et al., 2021). Building on this evidence, we derive our research hypotheses that focus on the interaction between the outcome bias and the decision to delegate choices to a human or an algorithm. We test whether delegation to the algorithm is pursued as an effective way to shift blame in case of undesired outcomes (Feier et al., 2022).

Given the evidence emerging from the literature, we pre-registered the following hypotheses. Hypotheses 1 and 2 refer to the Principal's behavior, while Hypotheses 3 and 4 refer to the Agent's behavior.

The first hypothesis we test concerns the impact of the Principal's knowledge of the investment outcome on the choice of t. We move from the evidence that individuals attach a positive value to good results, irrespective of the process leading to them and of the idiosyncratic risk preferences of the principal. In principle, rewards should depend on the Agent's ability to match the counterpart's risk preferences and exposure. However, Agents are not given access to information about their counterparts' risk preferences and thus cannot pursue a tailor-made investment strategy.

Hypothesis 1 (*Outcome Bias*)**.** Given an investment level, Agents obtain higher (lower) rewards in *Reward After* relative to *Reward Before* for positive (negative) investment outcomes.

The second hypothesis we test concerns the *shift in responsibility* when the Agent delegates the algorithm to invest. The algorithm's choice obscures the Agent's intentions and does not allow the Principal to formulate a causal link between intentions and consequences. This should dampen rewards/punishments in light of success/failure and, thus, reduce the size of the outcome bias.

[2] The Agents learn their reward only at the end of the experiment to avoid possible comparisons between their payoff and the payoff of the Principal. Beyond preventing strategic behavior and ensuring unbiased decisions, this will allow for more accurate measurement of preferences in the subsequent survey.

Table 1
Number of participants per condition and role.

Experiment	Condition	Principal	Agent
Preliminary		50	50
Main	Reward After	200	200
Main	Reward Before	201	200

Hypothesis 2 (*Outcome Bias and Delegation*)**.** Outcome bias is more prominent for Agents' firsthand choices than for choices made by the algorithm.

When the Agents anticipate the dampening in rewards/punishment associated with delegation, they may exploit delegation to the algorithm to "hedge" the risk associated with firsthand decisions in condition *Reward after*. Delegation in this condition seems particularly attractive for Agents who are averse to losses and put more weight on the punishment after a bad outcome (loss) than on the reward after a good outcome (gain) (Tversky & Kahneman, 1992).

Hypothesis 3 (*Delegation*)**.** Delegations to the algorithm are more frequent in *Reward After* than in *Reward Before*.

Finally, we test whether the Principal's knowledge of the investment outcome affects the risk-taking behavior of the Agent. Given our investment task, higher investment levels entail more risk and rewards. When rewards are conditional upon decisions and not outcomes, Agents may seek rewards based on expected results, given that outcomes are not yet accessible.

Hypothesis 4 (*Portfolio Composition*)**.** Agents undertake riskier investment on behalf of the Principal in *Reward Before* than in *Reward After*.

2.4. Procedures and participants

The study is an online pre-registered experiment, with English-speaking participants recruited via the Prolific platform. The experiment was programmed and conducted in oTree (Chen et al., 2016) on a private server and in English. The average duration of the experiment was 291 s and 311 s for Agents and Principals, respectively.

Choices in the experiment were made with tokens, with a conversion rate of 1 Token = 0.01 GBP. Participants received a fixed payment for participation, proportional to the estimated time to complete the experiment, and a bonus payment based on choices in the experiment. The fixed payment was 0.70 GBP (8.4/hr for an estimated duration of 5 min). The average bonus payment was 1.10 GBP for Principals and 0.70 GBP for Agents.

In each experimental condition, we recruited a gender-balanced sample of UK residents fluent in English. The total number of participants who completed the experiment is 901, distributed across alternative conditions as illustrated by Table 1. The *Preliminary* experiment serves as an input for the algorithm in the *Main* experiment. In *Preliminary*, Agents choose s and cannot delegate the choice to the algorithm. The Principal appropriates the investment returns, and the Agents are paid a fixed fee. The algorithm will randomly choose one of the Agents' choices as the investment choice s in the *Main* experiment. In the *Main* experiment, we implement the two controlled manipulations described in Section 2.2.

To avoid complications related to interaction in unsupervised online experiments, we adopted a "cold" matching procedure between choices of Agents and Principals over distinct sessions. Specifically, we first collected the Agents' choices and then matched each choice made by an Agent with a Principal. If a Principal withdrew from the experiment, an algorithm re-matched the Agent's choice with a new Principal. After the Principal choices were collected, participants' final payoffs were computed. All the information provided to participants was truthful, and the choices were anonymous.

Table 2
Investment levels in the two treatments (% of total).

Investment	Reward After	Reward Before
0	8	11.4
25	36	33.3
50	38	36.8
75	11	9.5
100	7	9.0

After completing the main part of the experiment, participants were asked to complete a survey about satisfaction regarding the outcome of the task results, punishment or reward actions, and trust in the Agent. After completing the survey, participants were told that their final payment (fixed + bonus) would be communicated on the Prolific platform. The computation of actual payoffs was performed offline, and participants were paid via Prolific.

3. Results

Here, we present the results of our experiment, following the structure of our research hypotheses. For the main results, we present descriptive statistics and the results of the statistical tests.

3.1. Outcome bias

Before focusing on the outcome bias, we test whether the two treatments are significantly different in terms of the investments chosen by the Agent. Table 2 shows the investment levels in the two treatments.

The average (median) investment level is equal to 43.250 (50) in *Reward After* and 42.786 (50) in *Reward Before*. A Wilcoxon rank sum test shows that the two distributions are not significantly different (p-value=0.780) in *Reward Before* and *Reward After*. The absence of differences across the two treatments is also confirmed by the Chi-squared test results (p-value=0.702). Regarding delegation choices, 50% and 51.7% of total choices are delegated to the algorithm in *Reward After* and *Reward Before*, respectively. A Chi-squared test confirms that the two distributions are not significantly different (p-value=0.803). Given the absence of differences in delegation and investment profiles across treatments, we present here the results of the analysis of the pooled sample of choices in *Reward Before* and *Reward After*.

Fig. 2 provides a representation of the distribution of reward choices by the Principal in condition *Reward Before* (central panel) and in conditions *Reward After*, when a success is registered (upper panel) and when a failure is registered (lower panel).

The average (median) reward in *Reward Before* is 20.597 (20). As the figure shows, the distribution of choices is skewed toward higher rewards in the case of success and toward lower rewards in the case of failure. The average (median) reward is 32.407 (30) in the case of success and 11.438 (10) in the case of failure. The central tendency of rewards is statistically different across the three conditions (Kruskal–Wallis rank sum test, p-value < 0.001). Furthermore, pairwise comparisons show that the average reward statistically differs in all comparisons between the three conditions (Wilcoxon rank sum test, all p-values < 0.001). This result provides evidence in support of Hypothesis 1 (Outcome bias).

Table 3 reports the results of a linear regression model, where the dependent variable is the reward and the independent variables are the experimental conditions and the investment levels. Column (1) presents a pooled regression, while column (2) contains the regression controls for the investment levels, and column (3) further controls for potential interactions between the experimental conditions and the investment outcomes.

The regression outputs in columns (1) and (2) show that individuals tend to reward Agents when a positive outcome is obtained (*Reward*

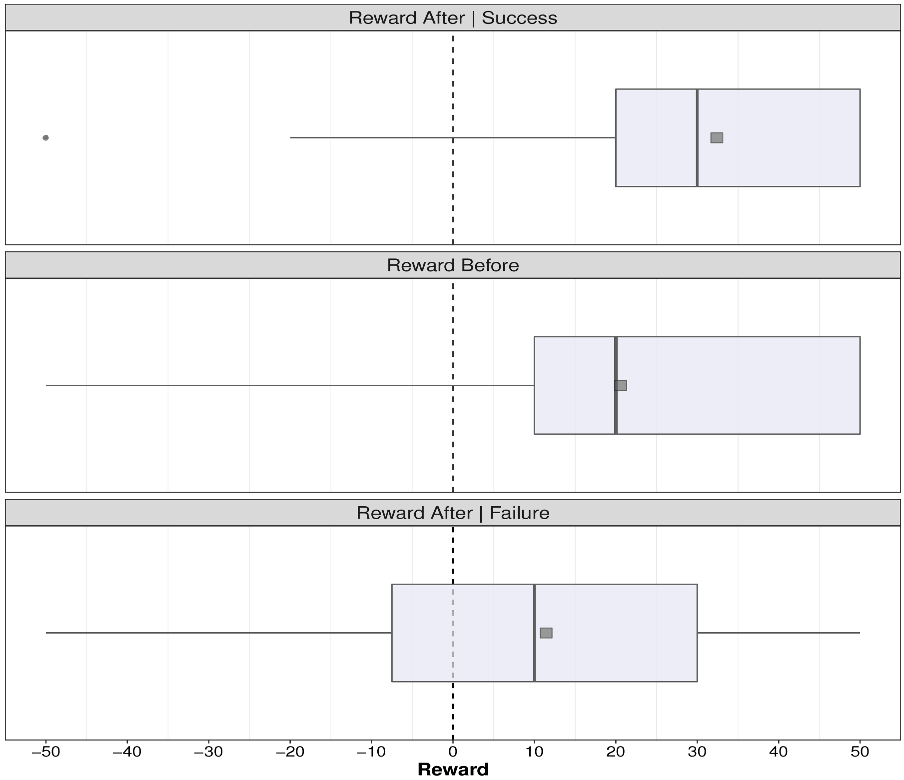

Fig. 2. Rewards across experimental conditions. Note: The square dot represents the average of the distribution, and the boxplot illustrates the conventional quartile partition.

Table 3
Outcome bias.

	(1)	(2)	(3)
(Intercept)	20.597 (1.728)***	21.832 (4.034)***	16.087 (5.059)**
Reward After|Success	11.810 (3.755)**	11.462 (3.748)**	28.913 (13.143)
Reward After|Failure	−9.159 (2.664)***	−8.942 (2.661)***	3.913 (8.640)
s=25		−0.562 (4.429)	8.092 (5.863)
s=50		0.900 (4.394)	5.129 (5.792)
s=75		−10.000 (5.468)°	−2.403 (7.522)
s=100		−4.846 (5.820)	1.691 (7.635)
Reward After|Success × s = 25			−28.648 (14.637)°
Reward After|Success × s = 50			−16.796 (14.326)
Reward After|Success × s = 75			−2.597 (18.732)
Reward After|Success × s = 100			−4.191 (18.778)
Reward After|Failure × s = 25			−16.611 (9.712)°
Reward After|Failure × s = 50			−8.783 (9.691)
Reward After|Failure × s = 75			−20.375 (11.761)°
Reward After|Failure × s = 100			−20.691 (12.892)
R2	0.072	0.089	0.118
Adj. R2	0.067	0.075	0.086
Num. obs.	401	401	401

Note: Results of a linear regression model. The dependent variable is the reward, and the independent variables are the experimental conditions and the investment levels. Significance levels: ***=0.001; **=0.01: *=0.05; °=0.1.

After|*Success*) and to punish them when a negative outcome is observed (*Reward After*|*Failure*), relative to the baseline condition in which the results are not known when choosing punishment/reward. The divergence in rewards is also confirmed by the Linear Hypothesis testing the equality between the two coefficients (*Reward After*|*Success=Reward After*|*Failure*, F-test, p-values<0.001). The regression outputs in column (3) confirm that successful investments encourage higher rewards than in the baseline condition. However, unsuccessful investments are not punished more than in the baseline condition. Furthermore, a Linear Hypothesis test shows that the coefficient of *Reward After*|*Success* is only marginally significantly different from that of *Reward After*|*Failure* (F-test, p-value=0.075). The interaction terms highlight the negative impact of moderate investment levels, both in case of success and failure. However, the interaction terms are only marginally statistically significant. Overall, the regression estimates of Table 3 show that the outcome bias is present, especially concerning rewards concerning successful investments.

Result 1 (*Outcome Bias*). *Successful investments are generally rewarded more than unsuccessful investments and investments whose outcome is not yet known.*

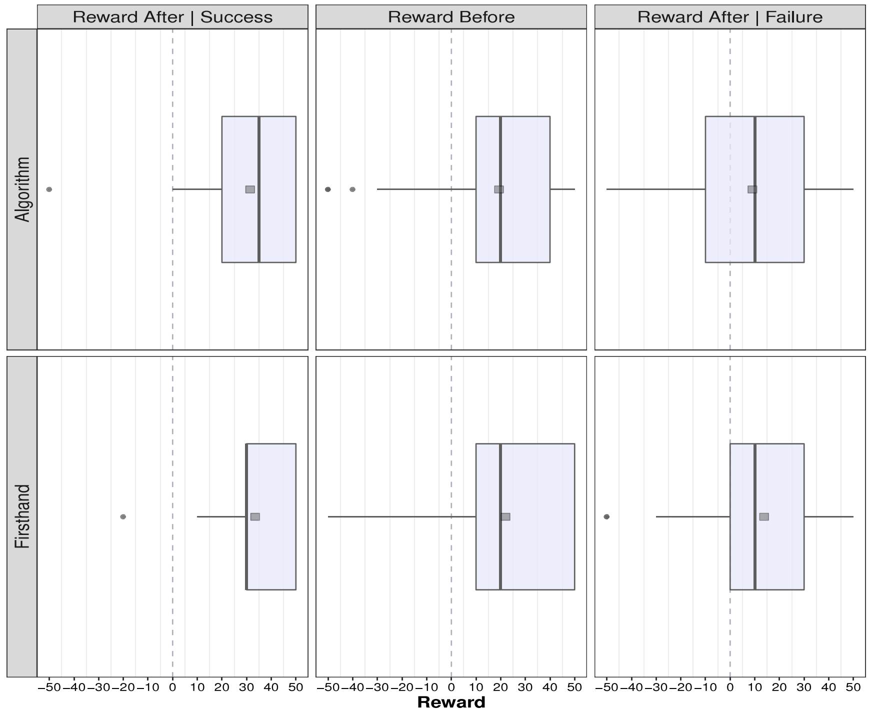

Fig. 3. Rewards by the Principal. Note: The square dot represents the average of the distribution, and the boxplot illustrates the conventional quartile partition.

3.2. Outcome bias and delegation

Fig. 3 provides a representation of the distribution of rewards by the Principal in condition *Reward Before* (central panel) and in *Reward After*, when a success is registered (leftward panel) and when a failure is registered (rightward panel). Differently than in Fig. 2, choices made by Agents firsthand and by the Algorithm, because of delegation, are kept separate (lower and upper panels, respectively).

The pattern highlighted in Fig. 2 is largely replicated by Fig. 3, with higher rewards after success. A significant difference across the three conditions is confirmed by a Kruskal–Wallis rank sum test, both for Firsthand choices (p.value < 0.001) and for Algorithm choices (p-value < 0.001).

Regarding our research Hypothesis 2, no significant differences are observed between Firsthand and Algorithm choices in each condition (Wilcoxon rank sum test, all p-values ≥ 0.220). Thus, we find no support for Hypothesis 2. This result is corroborated by the outcomes of the regression estimate reported in Table 4.

The regression outputs in column (1) confirm the existence of an outcome bias but no impact of delegation on rewards. Column (2) confirms that there is no significant interaction between the result of the investment in *Reward After* and the delegation choice. This also holds when controlling for different investment levels in column (3). Thus, we can state that:

Result 2 (*Outcome Bias and Delegation*)**.** *Outcome bias is present in choices made by the Agent firsthand and in choices delegated to the algorithm. Delegation to the algorithm does not significantly affect the size of the bias.*

3.3. Delegation

Table 5 reports the frequency of delegation choices in the two experimental treatments.

The frequency of Firsthand choice and delegation to the Algorithm is not different from 50% in both conditions (Proportion test, p-values ≤ 0.621). The lack of differences across conditions is confirmed by a Chi-squared test, which does not reject the null hypothesis of equal frequencies across the two conditions (p-value=0.764). Thus, we can safely state that the data do not support Hypothesis 3.

Result 3 (*Delegation*)**.**
Delegations to the algorithm are as likely as firsthand choices and are unaffected by the Principal's knowledge of the outcomes (Reward Before vs. Reward After).

3.4. Portfolio composition

The size of the investment share s defines the risk associated with a given investment. The higher the investment share, the higher the risk. Fig. 4 represents the investment distribution in the two experimental conditions for the investment made firsthand by the Agents.

The average investment level is nearly identical across conditions: 41.5 in Reward After and 41.4 in Reward Before. In both cases, the most frequent choice is 25, indicating low-risk exposure among Agents who retain control over the decision. Only a few observations align with the risk-neutral or risk-seeking investment level of 100. A chi-squared test fails to reject the null hypothesis of equal frequencies across conditions (p-value = 0.190). Therefore, our data do not support Hypothesis 4:

Result 4 (*Portfolio Composition*)**.** *Agents who make firsthand choices do not condition their investment on the Principal's knowledge of the outcomes (Reward Before vs. Reward After).*

3.5. Final questionnaire

After making their investment choice, all participants were asked to answer a questionnaire. The questionnaire has the same structure for the two roles: six questions with a Likert scale from "Strongly disagree" to "Strongly agree". However, the content of the questions was different for the two roles. The questionnaire for the Principal focused on the motivations to reward/punish the Agent, while the questionnaire for the Agent focused on the reasons to delegate to the algorithm and to choose a specific investment level. The outcomes of the questionnaire for the Principal are reported in Fig. 5, while outcomes for the Agent are reported in Fig. 6.

From the questionnaire, we can infer that most Principals agree with the idea that rewards should be conditional on the outcome of the

Table 4
Outcome bias and Delegation.

	(1)	(2)	(3)
(Intercept)	18.992 (2.091)***	19.327 (2.405)***	14.658 (5.430)**
Firsthand	3.325 (2.445)	2.632 (3.463)	2.529 (3.471)
Reward After|Failure	−9.239 (2.662)***	−10.299 (3.761)**	3.672 (8.971)
Reward After|Success	11.814 (3.751)**	12.102 (5.223)*	28.193 (13.723)*
Reward After|Success × Firsthand		−0.599 (7.525)	1.770 (7.608)
Reward After|Failure × Firsthand		2.124 (5.337)	2.483 (5.374)
s=25			8.125 (5.868)
s=50			5.568 (5.827)
s=75			−2.038 (7.544)
s=100			1.715 (7.641)
Reward After|Success × s = 25			−29.158 (14.667)*
Reward After|Success × s = 250			−16.876 (14.360)
Reward After|Success × s = 75			−2.962 (18.753)
Reward After|Success × s = 100			−3.141 (18.868)
Reward After|Failure × s = 25			−18.128 (9.795)°
Reward After|Failure × s = 50			−9.864 (9.731)
Reward After|Failure × s = 75			−21.296 (11.789)°
Reward After|Failure × s = 100			−21.049 (12.905)
R2	0.076	0.077	0.123
Adj. R2	0.069	0.065	0.084
Num. obs.	401	401	401

Note: Results of a linear regression model: the dependent variable is the reward, and the independent variables are the delegation choice, the experimental conditions, and the investment levels. Significance levels: ***=0.001; **=0.01: *=0.05; °=0.1.

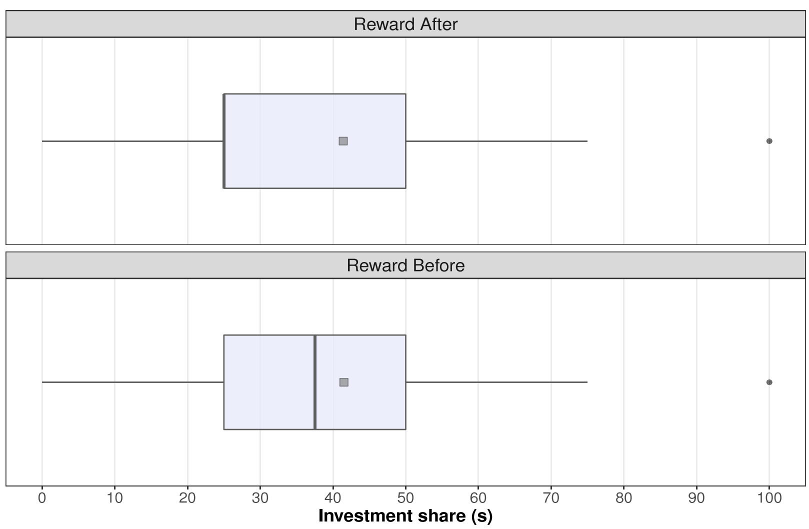

Fig. 4. Investment share in firsthand investments.
Note: The square dot represents the average of the distribution, and the boxplot illustrates the conventional quartile partition.

Table 5
Delegation choices.

Condition	Firsthand	Algorithm
Reward After	96 (48%)	104 (52%)
Reward Before	100 (50%)	100 (50%)

Note: Number of observations per condition with percentage frequencies within parentheses.

investment, in line with the outcome bias observed in the data. Furthermore, most Principals believe that Agents should feel accountable for the investment, even when they delegate to the algorithm. In this sense, the delegation does not seem to provide a "shield" against blame, following a negative outcome. This could explain why the outcome bias is not conditional upon delegation (see Result 2). Finally, there is no consensus among the Principals about the superiority of the algorithm over the human Agent and that a different delegation choice would have led to a different reward.

From the questionnaire, no consensus about the profitability of delegating to the algorithm emerges among the Agents. However, most Agents believe that the Principal would reward/punish differently before or after the results are known. This is in line with the outcome bias observed in the data. Only a minority disagrees with the idea that the risk of the investment could be higher when outcomes are not known to the Principal when choosing the reward. However, the dominant motivation guiding the decision to invest is to avoid losses, as testified by the low investment levels observed in the data.

4. Discussion and conclusion

The recent advent of generative artificial intelligence and large language models poses new challenges to understanding interactions between humans and algorithms. Finance seems to occupy a prominent place, with an increasing diffusion of robo-advising in financial decisions (Capponi et al., 2022). We focus here on choices in a controlled environment, capturing essential elements of fiduciary money management. A Principal delegates an Agent to invest resources on their behalf,

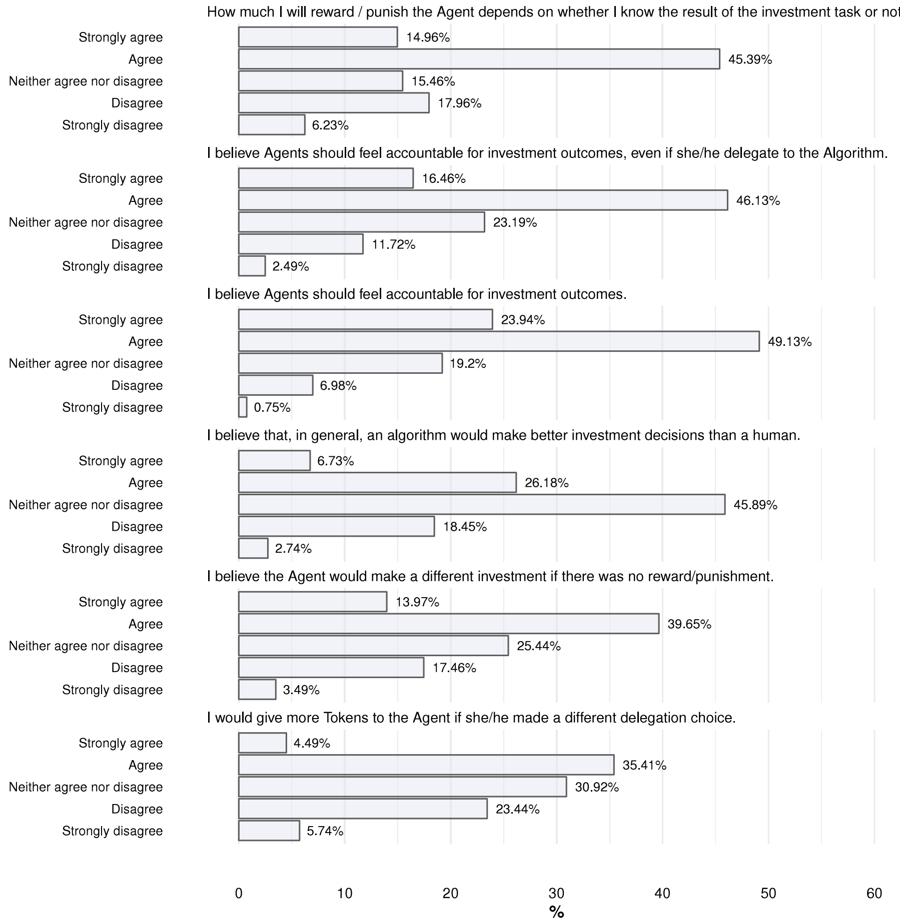

Fig. 5. Questionnaire Principal.

and the Agent can decide firsthand how to invest or, in turn, delegate an algorithm to do it. Within this setting, we investigate outcome bias, a well-documented phenomenon in which the quality of the decision-making process is evaluated ex-post mainly by its outcomes and not by its ex-ante validity.

Understanding how people perceive the relationship between choices and outcomes is crucial in situations where results do not follow deterministically from actions. For instance, consider an investor who, by sheer luck, outperforms the market despite having a poorly diversified portfolio. If decision quality is judged solely by outcomes, others may be encouraged to imitate this "successful" investor despite the underlying risk. This tendency is evident on social trading platforms, where traders who achieve substantial short-term gains often receive high visibility, attracting numerous followers (Pelster & Breitmayer, 2019). This pattern underscores how individuals are often influenced by positive outcomes, even when those results arise from luck rather than skill. As a result, seemingly successful but high-risk investment strategies may be replicated, reinforcing a cycle of speculative decision-making among followers.

Previous experimental works have documented the existence of outcome bias when financial choices are delegated to peers (König-Kersting et al., 2021). We contribute to this literature by showing that bias is also persistent when an algorithm is "hired" to choose, and this is publicly known. Those entrusting their resources to a fiduciary agent seem to perceive the results obtained by the algorithm as a direct emanation of the agent. On the one hand, agents may thus save on the cognitive and material cost of deliberately choosing investment strategies and still get rewarded for positive outcomes obtained by an algorithm. On the other hand, adopting the algorithm does not cause a "shift of responsibility" and, thus, does not provide a shield against bad outcomes. This result is especially relevant in a rapidly evolving context where algorithms offer significant automation opportunities, like finance and healthcare.

Throughout our investigation, we portray the asymmetry in rewards for successful and unsuccessful investments as originating in the outcome bias. However, alternative interpretations may be compatible with our results. For example, the observed phenomenon may reflect an application of an actualist utilitarian approach, in which the outcome, rather than the process or probabilities, serves as the primary determinant of value. This perspective challenges the straightforward labeling of the observed behavior as a "bias", which implies an irrational or improper deviation from a normative standard. Instead, the principals may have been consciously judging agents based on investment outcomes. Future research should aim to disentangle the influence of cognitive biases from conscious normative judgments in this context.

Our study serves as an initial exploration into understanding the role of algorithms in fiduciary money management and evaluation of investment outcomes. A limitation of our investigation lies in the specific nature of the adopted algorithm. Notably, our procedure lacks a generative aspect, and the algorithm's decisions are directly derived from humans' previous choices. However, it is important to stress that participants are unaware of the actual nature of the algorithm and, in this sense, face a condition similar to the one in the real world where the algorithm's inner workings are opaque.

The extent of autonomy bestowed upon the algorithm may significantly influence the relationship between outcomes and rewards.

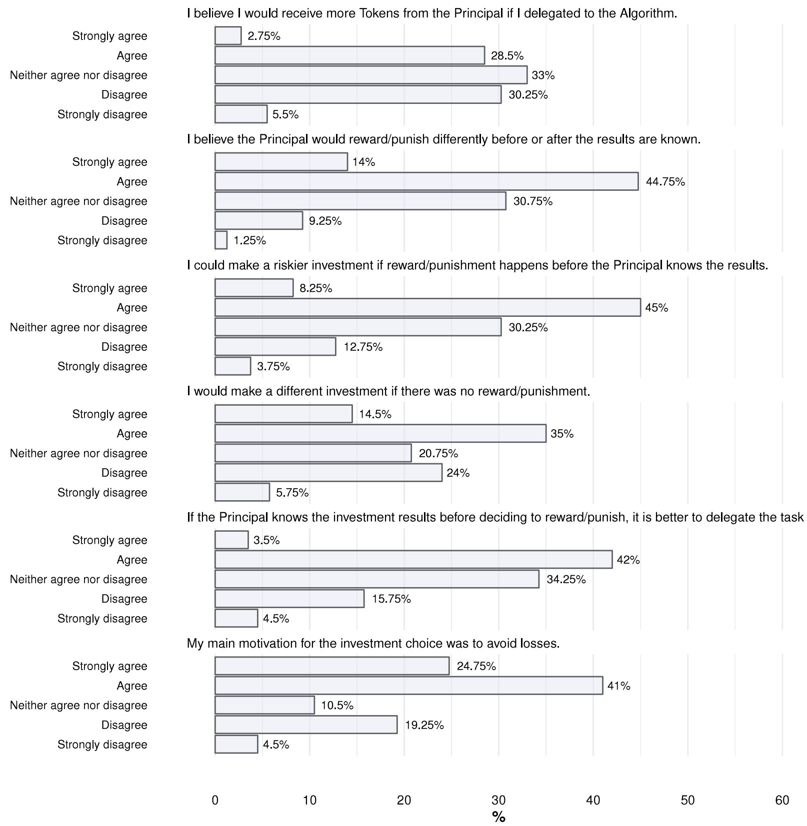

Fig. 6. Questionnaire Agent.

In our current approach, the algorithm's choices remain closely tied to human decisions. Future research endeavors could examine how varying degrees of autonomy might impact this linkage. Specifically, a heightened level of autonomy could introduce a greater separation between artificial intelligence and humans overseeing it, potentially complicating the assignment of responsibility for choices made by machines.

While our findings demonstrate that, in our specific context, humans do not merely "hide behind the machines" (e.g., Leib et al., 2021) to avoid potential retaliation, it is essential to recognize the possibility that this dynamic might emerge with more sophisticated algorithms. This consideration prompts the need for continued exploration into the nuanced interactions between human decision-makers and increasingly autonomous artificial intelligence systems.

During the preparation of this work, the authors used GPT-4o to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Zilia Ismagilova: Writing – review & editing, Writing – original draft, Visualization, Project administration, Formal analysis, Data curation, Conceptualization. **Matteo Ploner:** Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis, Data curation, Conceptualization.

Ethical Statement

As per the guidelines of the Ethical Committee of the University of Trento, formal ethics approval was not required as the study involved anonymous adult participants who faced no identifiable risk. The experiment was hosted on Prolific, where participants were informed of the study's nature, duration, and compensation. Participation was voluntary, and consent was obtained by virtue of participants opting to begin the study after reviewing the study specific information that was provided. Participants are free to withdraw at any time.

Declaration of competing interest

We declare no conflict.

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Update

Computers in Human Behavior: Artificial Humans

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Erratum regarding corrections related to Ethical approval and informed consent in articles published in [Computers in Human Behavior: Artificial Humans 4C (2025)]

Ethical approval and/or informed consent were not included in the published version of the following articles included in Volume 4C of Computers in Human Behavior: Artificial Humans.

The appropriate Ethical approval and/or informed consent statement, provided by the respective authors, are included below and have been added to the published version of the articles:

1. "What makes children perceive or not perceive minds in generative AI"[CHBAH (2025) 4C: 100135] https://doi.org/10.1016/j.chbah.2025.100135

Ethical approval and informed consent.

This study was approved by the Institutional Review Board of the University of Michigan (approval number HUM00227417). Children were recruited from two public libraries in a mid-west city in the U.S. between July and January 2024. Children participating in our study completed two sessions during story co-creation activities. Signed consent forms were obtained from parents or legal guardians, and verbal assent was obtained for children older than seven years of age. All children consented to participate in the study and were informed that they could discontinue at any time.

2. "A multinational assessment of AI literacy among university students in Germany, the UK, and the US"[CHBAH (2025) 4C: 100132] https://doi.org/10.1016/j.chbah.2025.100132

Ethical approval and informed consent.

This study was approved by the ethical committee of the Technical University of Munich (approval number 2023-251-S-KH). All included participants provided informed consent prior to data collection.

3. "Teaming up with robots: Analysing potential and challenges with healthcare workers and defining teamwork" [CHBAH (2025) 4C: 100136] https://doi.org/10.1016/j.chbah.2025.100136

Ethical approval.

This study was approved by the ethical committee of the German Psychological Society (TransMIT-Zentrum für wissenschaftlich-psychologische Dienstleistungen) (approval number Rosenthal-Von-DerPüttenAstrid2021-04-25VA).

4. "Quid pro Quo: Information disclosure for AI feedback in Human-AI collaboration" [CHBAH (2025) 3C 100137] https://doi.org/10.1016/j.chbah.2025.100137

Ethical approval.

The University of Hamburg uses an ethics questionnaire to assess the need for ethical review board approval. This study was deemed to be exempt from ethical review. The study was conducted ensuring voluntary participation, informed consent, and adherence to good ethical research practices.

5. "Enhancing emotional support in human-robot interaction: Implementing emotion regulation mechanisms in a personal drone" [CHBAH (2025) 4C: 100146] https://doi.org/10.1016/j.chbah.2025.100146

Ethical approval.

This study was approved by the institutional review board of the Ben-Gurion University of the Negev.

6. "Always check important information!" - The role of disclaimers in the perception of AI-generated content" [CHBAH, (2025) 4C: 100142] https://doi.org/10.1016/j.chbah.2025.100142

Informed consent was added to Section 2.1 Method as given below:

2.1. Method.

The study was approved by the ethical committee of the LeibnizInstitut fuer Wissensmedien in Tuebingen (LEK 2021/150). All participants in this study provided informed consent prior to data collection.

7. "Ain't blaming you: Delegation of financial decisions to humans and algorithms" [CHBAH (2025) 4C: 100147] https://doi.org/10.1016/j.chbah.2025.100147

DOIs of original article: https://doi.org/10.1016/j.chbah.2025.100161, https://doi.org/10.1016/j.chbah.2025.100136, https://doi.org/10.1016/j.chbah.2025.100146, https://doi.org/10.1016/j.chbah.2025.100142, https://doi.org/10.1016/j.chbah.2025.100147, https://doi.org/10.1016/j.chbah.2025.100141, https://doi.org/10.1016/j.chbah.2025.100148, https://doi.org/10.1016/j.chbah.2025.100163, https://doi.org/10.1016/j.chbah.2025.100132, https://doi.org/10.1016/j.chbah.2025.100137, https://doi.org/10.1016/j.chbah.2025.100149, https://doi.org/10.1016/j.chbah.2025.100160, https://doi.org/10.1016/j.chbah.2025.100135, https://doi.org/10.1016/j.chbah.2025.100152.

https://doi.org/10.1016/j.chbah.2026.100317

Available online 10 June 2026

Ethical Statement.

As per the guidelines of the Ethical Committee of the University of Trento, formal ethics approval was not required as the study involved anonymous adult participants who faced no identifiable risk. The experiment was hosted on Prolific, where participants were informed of the study's nature, duration, and compensation. Participation was voluntary, and consent was obtained by virtue of participants opting to begin the study after reviewing the study specific information that was provided. Participants are free to withdraw at any time.

8. "Evaluating the efficacy of Amanda: A voice-based large language model chatbot for relationship challenges" [CHBAH (2025) 4C: 100141] https://doi.org/10.1016/j.chbah.2025.100141

Informed consent was added in Section 4.2 Participants as given below:

4.2. Participants A total of 309 participants started the study. Of these participants, 49 did not complete the survey, 190 were not eligible due to the potential risk level being too high (e.g., reported suicidal ideation or potential domestic violence) or them not reporting on any specific relationship issue that they wanted to address (i.e., they selected "none" when asked about which issue they would like to address). A further 13 participants were excluded because they either did not interact or complete the interaction with the chatbot or had an issue with the chatbot interaction (their microphone was not working or there was too much noise in the background [e.g., participant was watching TV at the same time]). A further three participants did not complete the follow-up two weeks later (5.3 % dropout rate). Thus, the final sample consisted of 54 participants. Most participants were women (n = 34, 63.0 %), heterosexual (n = 44, 81.5 %), married (n = 27, 50.0 %), white (n = 205, 70.4 %), lived in the UK (n = 29, 53.7 %), and had at least an undergraduate degree (n = 39, 72.2 %). Most participants had also had at least some experience with chatbots (n = 46, 85.2 %) and/or virtual assistants (n = 49, 90.7 %). See Table 1 for the details of demographic variables for each group. All participants that were included in the study provided informed consent before participation.

9. "Socially excluded employees prefer algorithmic evaluation to human assessment: The moderating role of an interdependent culture [CHBAH (2025) 4C: 100152] https://doi.org/10.1016/j.chbah.2025.100152

Ethical approval.

This study was reviewed and approved by the Institutional Review Board of Sophia University (Approval No. 2023-133).

10. "Exploring the connecting potential of AI: Integrating human interpersonal listening and parasocial support into human-computer interactions" [CHBAH (2025) 4C: 100149] https://doi.org/10.1016/j.chbah.2025.100149

Informed consent was added in the 3.1.2. Procedure as given below:

3.1.2. Procedure.

The study received ethical approval from the University of Reading's School of Psychology and Clinical Language Sciences committee (2023-188-NW). All participants were provided with an information sheet and consent form and informed consent from all participants prior to data collection. Following procedures described in Itzchakov et al. (2020), participants were first asked to describe a time when they felt rejected, with the prompt: "Please spend a few moments thinking back to a time when you felt rejected by someone else or other people in your life." Then, participants interacted with Conversational Artificial Intelligence (AI) software (ChatGPT; GPT-3.5) in a back-and-forth conversation. Across six interaction points, the AI assimilated their responses during the conversation and responded as a listener (see coding below) while conveying either high-quality listening or lower-quality listening, based on assignment to condition. Participants responded in text boxes via their online survey, and then reported on the quality of the listening they received.

11. "Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams" [CHBAH (2025) 4C: 100148] https://doi.org/10.1016/j.chbah.2025.100148

Ethical approval.

This study was approved by the Institutional Review Board of Carnegie Mellon University (approval number STUDY2021_00000324).

12. "Promoting online evaluation skills through educational chatbots" [CHBAH (2025) 4C: 100160] https://doi.org/10.1016/j.chbah.2025.100160

Ethical Declarations.

No formal ethics committee approval was required for this type of study under German regulations and under the policies of the institution where the research was conducted. The study used an anonymous survey and interaction data, no collection of sensitive information, and no interventions that would trigger mandatory ethics review under German university guidelines.

All procedures adhered to the Ethical Principles of Psychologists and Code of Conduct of the American Psychological Association (APA, 2017) and to the ethical guidelines of the Deutsche Gesellschaft für Psychologie and the Berufsverband Deutscher Psychologinnen und Psychologen (DGPs and BDP, 2022). All participants provided informed consent prior to participation.

13. "Cognitive phantoms in large language models through the lens of latent variables" [CHBAH (2025) 4C: 100161] https://doi.org/10.1016/j.chbah.2025.100161

Ethics statement.

This study, procedure and data collection were approved by the local IRB before data collection (reference number TSB_RP1268). Informed consent was provided by all participants prior to commencement of data collection.

14. "Becoming dehumanized by a service robot: An empirical examination of what happens when non-humans perceive us as less than full humans" [CHBAH (2025) 4C: 100163] https://doi.org/10.1016/j.chbah.2025.100163

Ethical approval and informed consent.

This study did not require ethical board review by the Stockholm School of Economics. Participants in this study were recruited through the Prolific platform. Engagement with the platform and study selection is voluntary. Participants were provided with the study specific information prior to data collection and were free to withdraw at any time.

The published version of the following articles included in Volume 5C of Computers in Human Behavior: Artificial Humans contained text that was blinded for peer review and which had not been replaced upon publication.

The appropriate text, provided by the respective authors, are included below and have been added to the published version of the articles:

15. "Knowledge cues to human origins facilitate self-disclosure during interactions with chatbots" [CHBAH, (2025); 5: 100174] https://doi.org/10.1016/j.chbah.2025.100174

Three corrections where [ANONYMIZED FOR THE PEER REVIEW] text has been replaced with the exact text below:

3 Methods

The de-identified data sets, stimuli, and analysis code associated with this study are freely available online at https://osf.io/8vzwp/.

3.3 Ethics.

The research project received ethical approval from the University Ethics Committee (reference number 402210230).

3.4.1 The Chatbots.

Before the interaction with Sam, participants were introduced to it with the following text: "You are going to speak with Sam, a research assistant at the University of Glasgow.

16. "Gaze-informed signatures of trust and collaboration in human-autonomy teams" [CHBAH, 2025; 5: 100171] https://doi.org/10.1016/j.chbah.2025.100171.

The published article contain references to Location 1 and Location 2 text, this has been updated as follows:

2. Material and methods.

2.1. Participants.

A total of 83 volunteers from the University of Colorado, Boulder, and the United States Air Force Academy participated in the experiment. Participants were recruited through newsletter announcements and through the Sona online experimental recruiting software. Three participants were discarded due to technical difficulties with the system and 12 were removed due to poor eye tracking data quality (see Data Processing – Saccades, Fixations and Blinks below and Supplement A.1). As a result, data from 68 participants were used in all analyses. This final sample included 31 participants from the University of Colorado, Boulder (Age: M = 24.1 years, SD = 8; Gender: 20 female, 9 male, 1-non-binary/third gender, 1 preferred not to respond) and 37 participants from the United States Air Force Academy (Age: M = 19.2 years, SD = 1.9; Gender: 20 male, 17 female). Before participating, volunteers signed an informed consent document approved by the institutional review board at the United States Air Force Academy in accordance with the Declaration of Helsinki. Participants at the University of Colorado, Boulder were reimbursed with a $25 gift card while the United States Air Force Academy participants received extra course credit. All participants had normal vision (20/40 or better) without contact lenses. Three participants wore eyeglasses.